

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

Multi-Head Latent Attention + DeepSeekMoE

236B

Total Parameters

21B

Activated / Token

128K

tokens
Context Length

8.1T

Pretraining Tokens

The LLM Scaling Dilemma: Performance vs. Cost & Inference Speed

Challenge: Dense LLMs

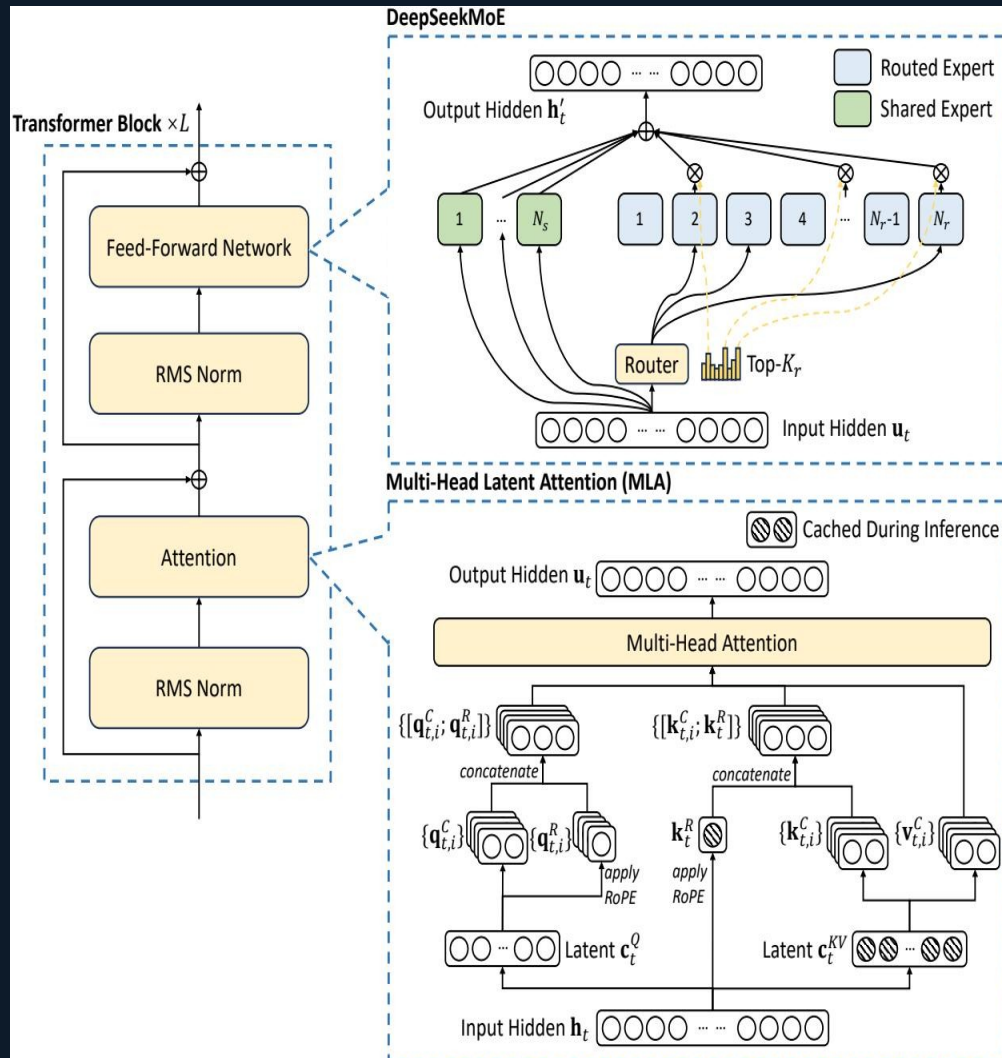
- ▶ Training 70B+ dense models costs tens of millions
- ▶ KV cache grows linearly with heads $\times$ layers $\times$ context
- ▶ Memory bandwidth bottleneck limits generation speed
- ▶ Long-context (128K) amplifies all of these problems

Why MoE + Efficient Attention?

- ▶ MoE decouples model capacity from activation cost
- ▶ Only a fraction of parameters activate per token
- ▶ Efficient attention reduces KV cache footprint
- ▶ Combined: larger capacity at lower compute cost

"Scale intelligence, not just parameters."

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in Each Transformer Block



Transformer Block Components

MLA (Multi-Head Latent Attention)

Replaces standard MHA. Compresses KV into latent vectors, reconstructed on-the-fly via learned projections. Decoupled RoPE for positional encoding.

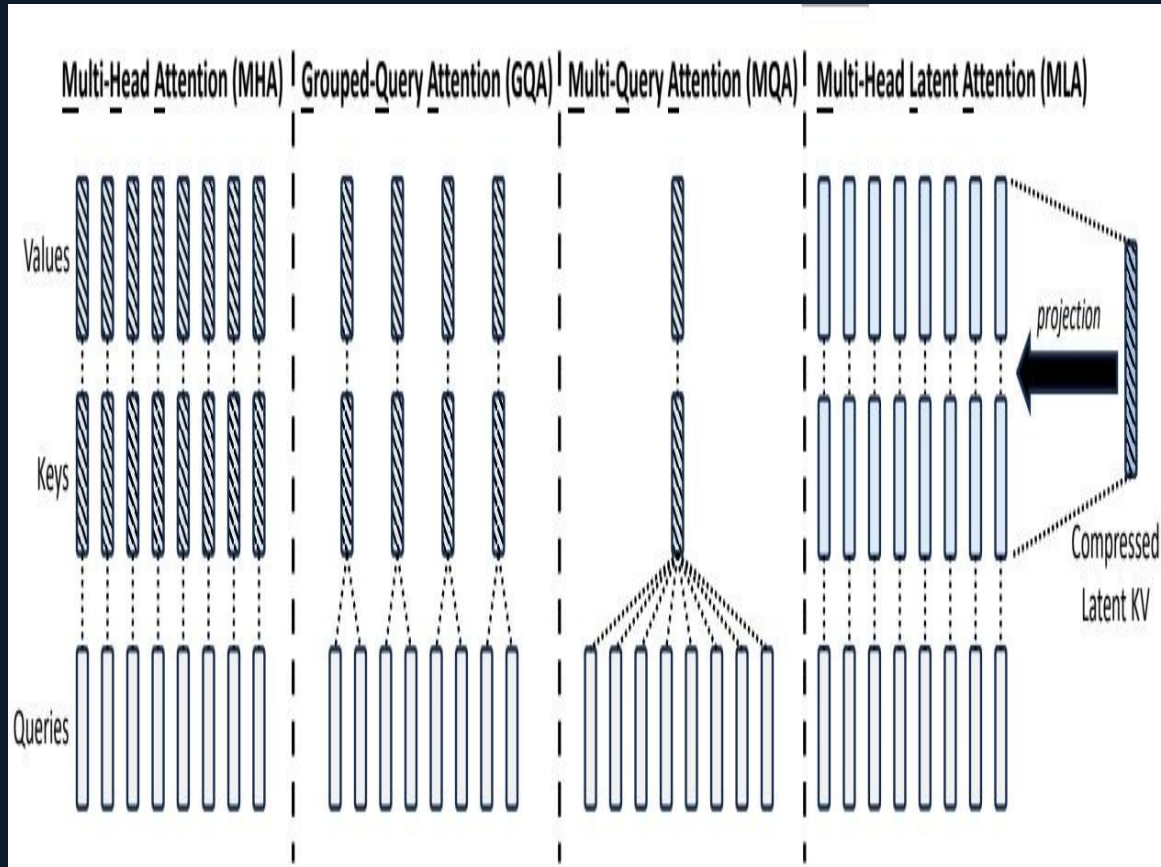
DeepSeekMoE (Feed-Forward Network)

Replaces dense FFN. Each token routes to shared + top-K routed experts. Device-limited routing minimises cross-node communication.

Model Scale

236B total parameters | 21B activated per token | 160 routed experts per MoE layer | 128K token context

Multi-Head Latent Attention (MLA): 93.3% KV Cache Reduction



How MLA Works

Low-Rank KV Compression

Instead of caching full K and V vectors per head, MLA compresses them into a single compact latent vector c_t^{KV} .

On-the-Fly Reconstruction

Full keys, values, and decoupled RoPE positional keys are reconstructed via learned projection matrices—not stored.

Outperforms MHA

MLA achieves better accuracy than standard Multi-Head Attention while using a fraction of the KV cache.

vs. GQA / MQA

Unlike GQA and MQA (which degrade quality to save cache), MLA improves quality AND reduces cache by 93.3%.

93.3% KV Cache Reduction vs. standard MHA

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

Always activated. Capture universal, cross-domain knowledge present in every token.

Routed Experts (N_r)

Sparsely activated via top- K_r routing. Each expert learns highly specialised representations.

Fine-Grained Segmentation

Experts are finer-grained than prior MoE designs, raising specialisation potential and capacity utilisation.

Device-Limited Routing

Each token is constrained to experts on at most M devices, eliminating all-to-all cross-node communication overhead.

Auxiliary Losses

Load-balancing losses enforce uniform expert utilisation. Token-dropping during training prevents expert overflow at inference.

Training Configuration

8.1T

Pretraining tokens

1.5M

SFT conversational sessions

GRPO

RL algorithm (Group Relative Policy Opt.)

160

Routed experts per MoE layer

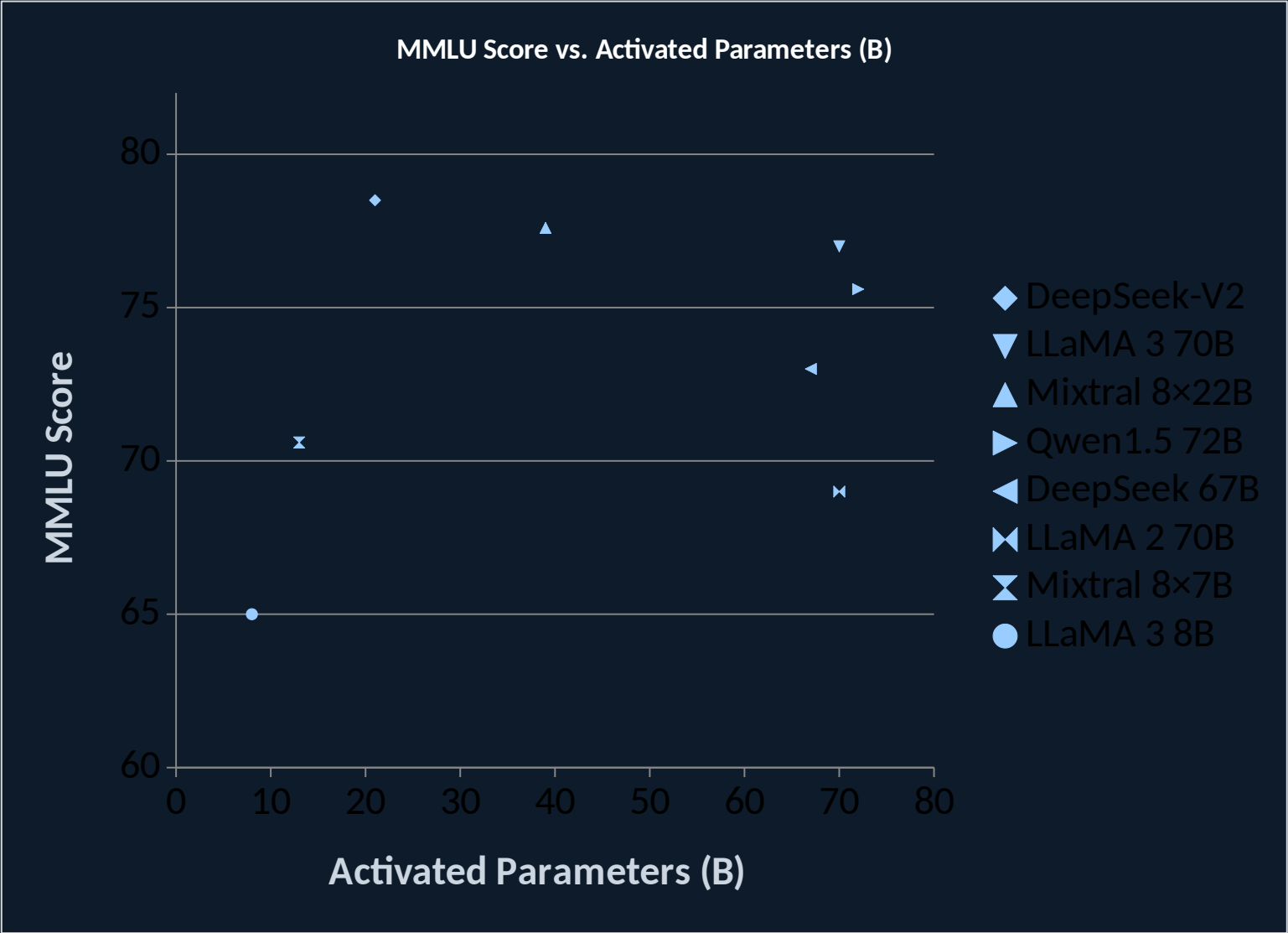
2

Shared experts per MoE layer

42.5% Training Cost Reduction and 5.76× Throughput vs. DeepSeek 67B

DeepSeek 67B (Dense Baseline)	DeepSeek-V2 (MoE Innovation)
Training Cost Baseline (1×)	Training Cost 42.5% lower
KV Cache Size Baseline (1×)	KV Cache Size 93.3% smaller
Max Gen. Throughput Baseline (1×)	Max Gen. Throughput 5.76× faster
Activated Params 67B parameters	Activated Params 21B parameters
Total Params 67B parameters	Total Params 236B (capacity)

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters (Pareto-Optimal)

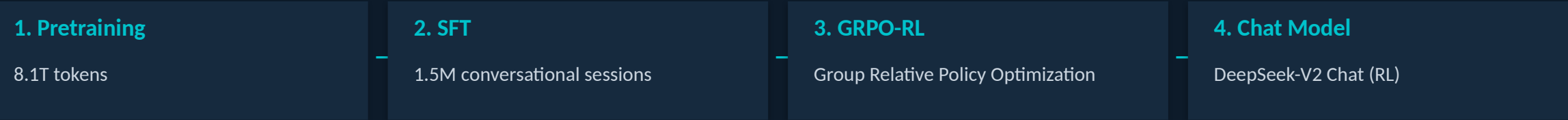


Model Comparison

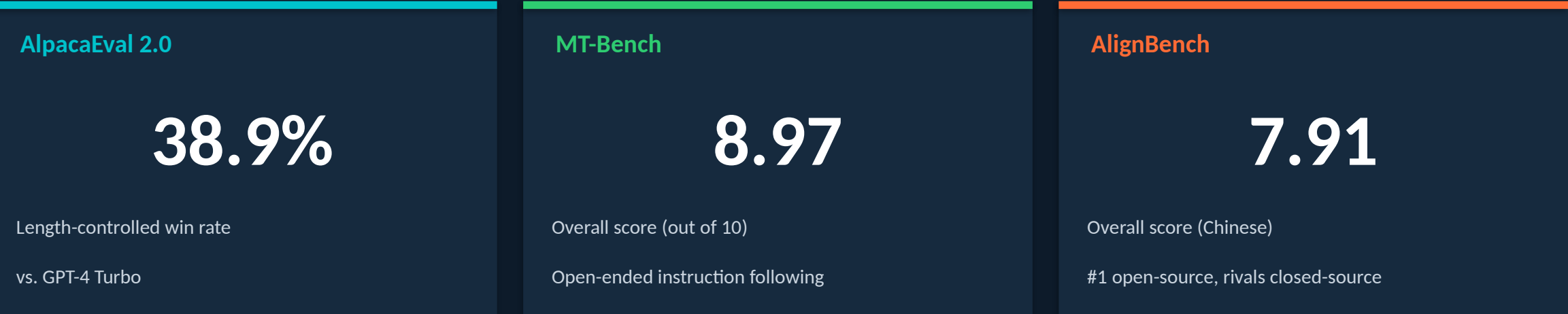
Model	Act.Params	MMLU
DeepSeek-V2	21B	78.5
LLaMA 3 70B	70B	77.0
Mixtral 8×22B	39B	77.6
Qwen1.5 72B	72B	75.6
DeepSeek 67B	67B	73.0
LLaMA 2 70B	70B	69.0
Mixtral 8×7B	13B	70.6
LLaMA 3 8B	8B	65.0

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chat Benchmarks

Post-Training Pipeline



Chat Evaluation Results



DeepSeek-V2 Chat (RL) is the strongest open-source model in Chinese and competitive with closed-source systems in English.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

MLA is a Strict Improvement over MHA

Low-rank KV compression achieves better accuracy than MHA, GQA, and MQA while reducing KV cache by 93.3% — the right choice for long-context inference.

DeepSeekMoE Makes 236B Affordable

Fine-grained shared + routed experts with device-limited routing cut training costs 42.5% vs. a 67B dense model while dramatically increasing capacity.

21B Activated Params Outperform 70B Dense Models

DeepSeek-V2 surpasses LLaMA 3 70B and Qwen1.5 72B on MMLU with 3× fewer activated parameters — a clear Pareto-optimal result.

5.76× Generation Throughput Enables Real Deployment

Higher throughput makes DeepSeek-V2 practical at scale. KV cache reduction directly translates to lower serving costs and higher concurrency.

State-of-the-Art Multilingual Alignment

GRPO-based RL produces the strongest open-source Chinese chat model, competitive with proprietary systems on English benchmarks.